

Global Satellite Broadband: A Deployment Guide

Executive Summary: Global broadband via satellite (e.g. Starlink, OneWeb, Kuiper) is enabled by large Low-Earth-Orbit (LEO) constellations offering low latency and high throughput beyond GEO/MEO. Key architectural trade-offs include LEO's sub-2,000 km orbits (tens of milliseconds delay) versus MEO/GEO's higher latency [17†L181-L189] [17†L190-L198]. Constellation design (number of satellites, altitudes, orbital planes and phasing) is driven by coverage and capacity needs. For example, Starlink plans multi-shell fleets (~6,000+ sats, ~340–570 km [2†L69-L75]), OneWeb deployed 648 sats at ~1,200 km [64†L608-L616], and Kuiper targets 3,236 sats at 590–630 km [2†L103-L107]. Satellites use flat-panel phased-array antennas and Ka-/Ku-band links for user access [17†L197-L204] [64†L625-L630]; many now include high-speed optical inter-satellite links (ISL) to form a mesh network [2†L79-L87] [64†L618-L622]. Ground segment comprises widely distributed gateways (for connecting space to the terrestrial internet) and electronically steerable user terminals with low-cost phased-array antennas (e.g. Starlink's flat "Dishy" terminals) [3†L37-L40]. The network architecture adapts terrestrial routing and mobility techniques: satellites/ground gateways use dynamic routing and handovers as users track moving beams, with Quality of Service (QoS) managed via traffic prioritization and bandwidth allocation.

Spectrum and regulation are critical: LEO systems use FSS allocations (Ka/Ku bands and emerging V/Q bands) coordinated via ITU and national authorities. WRC-19 set strict "use-it-or-lose-it" milestones (10% of satellites deployed in 2 years, 100% in 7) [39†L129-L137] to prevent spectrum warehousing. Operators must register with ITU and obtain country licenses (FCC in the US, Ofcom in the UK, etc.), complying with radio-power limits to protect GEO satellites. Key bands include Ka-downlink (~17–20 GHz)/uplink (~27–30 GHz) and Ku-downlink (~10–12 GHz)/uplink (~14–14.5 GHz) (exact sub-bands per license). National regulators may impose additional requirements (e.g. UK's security review for OneWeb, spectrum fees, terminal certification).

Manufacturing and launch are capital-intensive. Satellites (hundreds of kilograms, with batteries, processors, and antennas) are now mass-produced (e.g. Airbus/OneWeb JV, SpaceX's Redmond factory) at ~\$0.2–0.4M each [56†L478-L481]. LEO sats typically carry dozens of beams, advanced RF/optical payloads, and life expectancy ~5–7 years [17†L206-L212]. Launch options range from SpaceX Falcon 9 (~60–100 sats per flight) to Ariane/Soyuz/Vulcan/New Glenn, with costs roughly \$0.5–2.0M per satellite [56†L490-L499]. For example, SpaceX's ~28 M\$ Falcon9 launches (60 sats) imply ~\$0.5M/sat, whereas one analysis assumed Kuiper's rockets cost ~\$1.5M/sat and OneWeb's ~\$2.0M/sat [56†L490-L499]. Rideshare and reusable rockets (Falcon 9, soon Starship) are lowering costs. A tentative CAPEX model:
$$\text{CAPEX} \approx \$(\text{\#sats}) \times (\text{sat cost} + \text{launch cost}) + \text{ground capex} + \text{spectrum fees}$$
 (often hundreds of millions) [56†L470-L478] [56†L490-L499]. For instance, 5,000 satellites at \$0.75M each (sat+launch) would cost ~\$3.75B, plus ~\$100M for gateways. OPEX (network operations, staff, insurance) runs hundreds of millions per year; one study assumed ~\$60M/yr staff and \$50M/yr marketing [56†L474-L478]. Marginal cost per bit falls rapidly with scale, but per-user cost remains high if many users share limited satellite capacity [15†L69-L77].

System Architectures: Satellites may be placed in LEO, MEO, or GEO, each offering distinct trade-offs. LEO constellations (400–2,000 km) deliver round-trip latencies <50 ms [17†L190-L198], enabling low-latency broadband (comparable to terrestrial). They require thousands of sats and continuous coverage (Figure 1). MEO orbits (~8,000–12,000 km, e.g. SES's O3b) have ~125–150 ms latency and need tens of sats, balancing coverage with reduced handovers. GEO systems (36,000 km) cover ~1/3 Earth per sat

with ~500–600 ms delay, suitable for broadcast/bulk data but too latent for real-time apps 【17†L181-L189】 . Compared to GEO/MEO, LEO sats are lower-cost per unit and can use high-frequency bands (e.g. Ka/Q/V with >20 GHz total bandwidth) for huge aggregate capacity 【17†L197-L204】 , but at the cost of much shorter orbital life (~5 yr) 【17†L206-L214】 and complex tracking. The table below summarises key attributes:

Orbit	Altitude	Latency (RTT)	# Sats for Global Coverage	Coverage per Sat	Pros	Cons
LEO	400–2,000 km 【17†L181-L189】	~20–50 ms	Thousands (e.g. 4–12k) 【2†L69-L75】 【64†L608-L616】	Tens of km radius	Very low latency; high link budget (short path); new high-bandwidth tech (Ka/Q/V) 【17†L197-L204】 ; easy capacity scaling 【17†L202-L210】	Many sats needed; rapid handovers; doppler shifts; higher station density; orbital decay (short life) 【17†L206-L214】
MEO	~8,000–12,000 km	~100–150 ms	100–200	~100s km	Moderate latency; fewer sats; still global coverage (e.g. O3b)	Higher latency & path loss than LEO; more costly sats; legacy systems limited
GEO	~36,000 km 【17†L181-L189】	~500–600 ms	3 (for full Earth)	~1/3 Earth	Fixed coverage, stable orbits; mature tech	Very high latency; large gateway antennas; limited to lower freq (C/Ku) due to rain fade; not polar coverage

【Table: GEO vs MEO vs LEO broadband satellite trade-offs.】

Constellation Design: Key design parameters are altitude, orbital planes, inclination, and phasing. Polar or near-polar orbits ensure global reach; Starlink, for example, uses multiple shells including polar and inclined, whereas Kuiper targets 30°–51.9° inclination (focusing mid-latitudes) 【70†L92-L100】 . Inclination choices determine coverage; high inclinations serve polar regions (as OneWeb’s 86.4° polar shells 【64†L608-L616】), whereas lower inclinations cover dense mid-latitudes with fewer sats. Constellation size is driven by capacity goals: more satellites allow smaller coverage cells (higher spectral reuse) and more per-user throughput. Satellite phasing must be arranged so that as one

satellite sets below horizon, another rises with overlapping coverage. For example, OneWeb's 648 satellites are arranged in 12 planes of 49 each at ~1,200 km [64†L608-L616] . Starlink's hundreds of planes (dozens per shell) ensure near-continuous global handover [2†L69-L75] . Orbital shell altitude also trades latency for coverage: higher orbits need fewer sats (as with OneWeb's 1,200 km design) but incur ~40 ms extra round-trip delay vs ~20 ms for low LEO [3†L49-L57] . Phasing between planes must be carefully chosen (e.g. evenly offset) to balance load and minimize simultaneous satellite omissions.

Satellite Payloads: Modern LEO broadband sats carry multi-beam phased-array antennas, transponders in Ku/Ka (and possibly Q/V) bands, and increasingly on-board processing and inter-satellite lasers. For instance, Starlink v1 sats use Ku/Ka for user links; v2 ("Mini") adds Ka and Ku transceivers plus E-band laser ISLs [2†L79-L87] . OneWeb Gen1 sats (~150 kg) support user links in Ku-band and gateway links in Ka-band [64†L625-L630] (no ISL); next-gen OneWeb will add higher throughput and possibly ISLs. User beams are digitally steered to cover small ground cells, often hundreds to thousands of spots per satellite. RF amplifiers and beamforming chips determine link budgets; typical LEO sat antennas have gains ~30–40 dBi. Growth areas include optical inter-satellite links: Starlink and Kuiper plan full OISL meshes to route traffic in-space [2†L79-L87] , reducing ground latency (light in vacuum is ~1.4× faster than in fiber [70†L112-L115]). Payload power is supplied by deployable solar panels (flat arrays or wings) and battery packs. Lifespan is ~5–7 years, with deorbit propulsion for end-of-life disposal (e.g. OneWeb complies with 25-year re-entry guideline [64†L625-L630]).

Ground Segment: The ground network has (a) **Gateways** to link satellites to the internet backbone and (b) **User terminals** for consumers/enterprises. Gateways use large antennas (often 3–10 m dishes or phased arrays) at sites with good sky visibility and fiber backhaul. Without ISLs, many gateways are needed to maintain low-latency paths; with mesh ISLs, fewer ground points suffice (e.g. Starlink plan is <200 gateways globally). Terminals are flat-panel phased arrays (~30–60 cm) that electronically track moving satellites. SpaceX has driven down Starlink dish costs to a few hundred dollars [3†L37-L40] ; Amazon, OneWeb and others pursue similar low-cost phased arrays. Terminals must handle Doppler shifts and fast handovers (switching beams as satellites move) without service drop. Both gateways and user equipment require RF front-ends (LNA/SSPA), digital processors, and position/attitude sensors. Tracking and control of ground antennas (for gateways) is simpler (they track slower LEO passes or use phased arrays).

Network Architecture: The LEO broadband network can be viewed as a dynamic mesh of space and ground links. Satellites interconnect via ISLs (optical or Ka-/Ku-band) and/or via gateways. Routing of IP traffic can occur onboard (satellite switches/routers) or in gateways. In fully meshed constellations, data can be forwarded "space-only" (e.g. Alaska ↔ Europe via path through satellites), reducing hops. Routing algorithms are specialized for the rotating geometry; possible solutions include dynamic time-dependent routing or delay-tolerant protocols (DTN) when needed. Session continuity ("handover") is akin to cellular: as one sat sets, the terminal transitions to the next, requiring synchronization of network state. Latency remains low for local hops, but end-to-end delay also depends on ground segments and inter-continental fibre (though optical ISLs can outperform some fibre routes due to faster light speed in vacuum [70†L112-L115]). QoS is managed by traffic scheduling (guaranteeing voice/enterprise traffic ahead of bulk data) and by capacity allocation in each beam. Peering arrangements connect gateways to the global internet; some constellations leverage parent company networks (e.g. Kuiper via AWS) to optimize routes.

Spectrum & Regulation: LEO broadband uses FSS (Fixed-Satellite Service) bands allocated by ITU: typically **Ku-band** (down: ~10.7–12.7 GHz, up: ~14–14.5 GHz) and **Ka-band** (down: ~17.7–20.2 GHz, up: ~27.5–30 GHz). Newer proposals include **V-band** (40/50 GHz) for ultra-high capacity. These bands are

licensed nationally; e.g. Starlink's FCC filings specify operations in the US Ka/Ku allocations. Non-GSO systems must coordinate to avoid interfering with GSO networks (power flux-density limits). The 2019 ITU WRC created a milestone-based regime: LEO fleets must deploy 10% of their filed constellation within 2 years, 50% in 5, 100% in 7 [39†L129-L137] ; this "use-it-or-lose-it" rule curbs speculative filings. Operators must register frequency assignments with the ITU Master Register and negotiate with other administrations (especially if coverage spans borders). National regulators (FCC, Ofcom, ANACOM, etc.) impose additional criteria: e.g. domestic licensing for gateways, spectrum fees, and foreign ownership reviews. The UK's stake in OneWeb illustrates strategic controls: some governments may mandate local production or data routing. Compliance obligations include debris mitigation (e.g. deorbit within 5–7 years post-service) and, in some cases, "light pollution" limits (Starlink uses visors to dim reflections).

Manufacturing & Launch: Modern constellations rely on high-rate satellite manufacturing (tens per week) and bulk launch campaigns. SpaceX builds Starlinks on a production line at ~0.25M\$ per unit [56†L478-L481] . OneWeb partnered with Airbus for two-per-day assembly. Launch contracts are a mix of dedicated and rideshare: SpaceX (Falcon 9, Starship) is Starlink's main launcher, reducing cost via reuse. OneWeb used Arianespace (Soyuz, Ariane 6) and India's GSLV Mk3 to launch >600 sats [64†L649-L657] . Amazon Kuiper has contracts with ULA Vulcan, Blue Origin New Glenn, and Ariane 6. Launch economics dominate: one analysis estimated ~\$0.5M launch cost per Starlink sat (based on \$28M Falcon9 with 60 sats) vs \$1.5–2.0M for Kuiper/OneWeb rockets [56†L490-L499] . Decreasing launch cost (e.g. SpaceX's <2k\$/kg Falcon-9 reusable) is critical to constellation viability. Ground infrastructure build-out (gateways, mission ops) is another major component.

Operations: Once deployed, operations include network NOC/MOC (monitoring sat health, scheduling maneuvers) and customer support. Telemetry, Tracking & Control (TT&C) links from each sat are managed by mission control centers. Collision avoidance is crucial: with thousands of LEO sats, operators rely on space surveillance data (e.g. US Space Force tracking) to predict conjunctions. Autonomous collision-avoidance maneuvers are routine (SpaceX reports performing dozens of avoidance burns monthly). Debris mitigation: operators plan controlled deorbit or graveyard orbits. Many (Starlink, OneWeb) pledge end-of-life disposal within ~5 years to limit space junk [17†L206-L214] [64†L625-L630] . Ground systems must also maintain high reliability, with backups for data links and power. Ongoing R&D improves satellite capabilities (e.g. enhanced payloads) through software updates or successive launches.

Business Models & Pricing: Broadband constellations generally sell tiered subscription plans. Starlink's model (as of 2026) offers residential service (in rural areas) for ~\$90–130/month plus a one-time ~\$500 terminal fee, business plans ~\$500/month, and premium mobility plans (aviation/maritime) at higher prices. Amazon Kuiper is expected to target similar or lower pricing, possibly bundling with Amazon Prime/AWS. OneWeb focuses on enterprises, telcos and governments; service is often sold as wholesale capacity or multi-year contracts (e.g. rural ISP backhaul, government networks) rather than direct consumer plans. Value-added markets include aviation and maritime connectivity at premium rates. Pricing strategies vary by region and competition; for example, in emerging markets satellite broadband may undercut limited terrestrial options, whereas in wealthy regions it must justify cost vs fibre/5G. Governments may subsidize household terminals (as with the Australian NBN satellite program) to accelerate uptake. Revenue per user (ARPU) must exceed the significant network cost per-user; one analysis finds LEO networks are viable only if user density remains low ($\approx 0.1\text{--}0.5$ users/km²) [15†L69-L77] . Marginal cost includes terminal hardware, spectrum fees, and incremental satellite/O&M costs; volume discounts (sat mass manufacturing, launch economy of scale) help reduce per-user costs.

CAPEX/OPEX & Economics: CAPEX is dominated by satellites and launch. Table 1 (below) illustrates a notional breakdown (based on [56]) for Starlink-class vs smaller constellations:

Cost Item	Starlink (6k sats)	Kuiper (3.2k)	OneWeb (648)	Type
Satellites (capex)	6,000×\$0.25M = \$1.5B	3,200×\$0.25M = \$0.8B	648×\$0.25M = \$0.162B	CapEx
Launch (capex)	6,000×\$0.5M = \$3.0B	3,200×\$1.5M = \$4.8B	648×\$2.0M = \$1.3B	CapEx
Gateways + Infra	~\$100M–200M	~\$50M–100M	~\$50M	CapEx
Spectrum Licenses	~\$125M (one-time)	~\$125M	~\$125M	CapEx
Total CapEx	~\$4–5B+	~\$5–6B	~\$1.7–2B	
Annual Staff/OPS (OPEX)	~\$60–100M	~\$30–60M	~\$10–20M	OpEx
Other O&M (e.g. insurance, power)	~\$50M/yr	~\$30M/yr	~\$5M/yr	OpEx
Marketing/Support	~\$50M/yr	~\$30M/yr	~\$10M/yr	OpEx

Table 1: Illustrative CAPEX/OPEX for LEO broadband systems (estimates). Data partly adapted from [56] and industry reports.

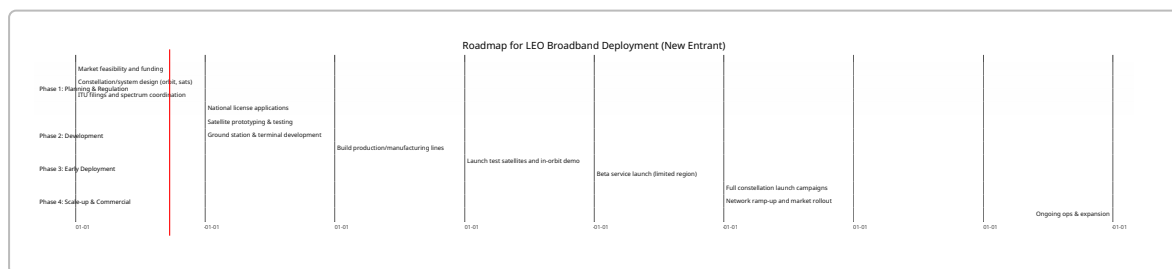
Financial viability requires balancing this high upfront cost against subscriber growth. For example, 5,000 sats (\$3.75B sat+launch) serving 2M users would need ~\$1,875 capex per user, amortized over ~5 yr ≈\$375/yr, plus OPEX. In practice, revenue per user is << this, so scaling to many millions of users (as Starlink aims) and keeping costs low are essential. Sensitivity: even a small increase in launch or satellite cost greatly raises CAPEX (doubling launch cost doubles budget). Conversely, terminal cost reduction and organic traffic (sideloading on terrestrial networks) can improve margins. Partnership models (leasing capacity to telcos or government) can also offset costs. Industry analyses suggest LEO broadband may eventually achieve per-user costs comparable to terrestrial backhaul in remote regions [17†L202-L210] [15†L69-L77], but only with massive scale and high ARPU services.

Competition: The primary global players are **SpaceX Starlink**, **Amazon Kuiper (Leo)**, **Eutelsat OneWeb**, and **Telesat Lightspeed**. SpaceX is by far the leader: >6,000 sats deployed (planning ~12k), >3 million subscribers [2†L69-L75], multi-orbit coverage (polar and mid-latitudes), and an integrated supply chain (own rockets and manufacturing) [2†L69-L75] [70†L91-L99]. Amazon Kuiper (deep-pocketed challenger) has FCC clearance for 3,236 sats in multi-shell LEO [2†L103-L107], with initial service started in 2025. Kuiper’s edge is AWS integration (edge cloud) and strong consumer distribution; its architecture uses fewer sats than Starlink but relies fully on OISLs from the start [2†L109-L117] [70†L86-L94]. OneWeb (backed by Eutelsat, UK government, Bharti) completed 648-sat LEO deployment (1,200 km) and targets enterprise/government markets rather than mass retail [53†L21-L30]. OneWeb’s first-gen sats lack ISLs [64†L618-L622], so it uses many gateways; a second-gen lower-altitude design with lasers is planned but delayed. Telesat (Canada) aims 298 sats for high-end enterprise/military, with full optical ISLs [2†L142-L149], but financing and deployment are still in flux. **Regional players:** China (Hongyan/Tianyan consortia) and India (Reliance Jio, 13,000-sat plan) have announced LEO projects, largely national in scope; SES’s O3b (MEO) and Hughes/Viasat (GEO) remain in niche markets (e.g. enterprise, remote cells). Incumbent GEO operators (Intelsat, Eutelsat, Viasat) are

investing in LEO to hedge. The competitive landscape implies “winner-takes-most” in consumer broadband (due to terminal stickiness) [2†L161-L169] , while specialized segments (government, IoT) may sustain multiple providers.

Risks & Mitigation: *Technical:* The systems are complex. Delays or failures in manufacturing or launch (e.g. a launch failure destroying many sats) can derail schedules. Mitigations: use proven rockets, iterative testing (e.g. launching small test sats first), and rapid iteration of satellite design. *Regulatory:* Spectrum coordination is fraught; failing WRC milestones or incurring interference disputes can force operations to pause. Mitigation: conservative filings, front-loading launches to meet milestones [39†L129-L137] , and legal teams for international coordination. *Financial:* The need for multi-billion-dollar investment before revenue poses cash-flow risk (OneWeb's 2020 bankruptcy is a cautionary example). Mitigation: staged funding (debt/equity tranches tied to deployment milestones), anchor customers or government backing (as with UK/Bharti's OneWeb rescue), and revenue diversification (e.g. selling wholesale capacity). *Market:* If demand (esp. consumer uptake) is weaker or slower, ARPU may not cover costs. Mitigation: targeting premium/enterprise markets, bundling with terrestrial services, and gradually expanding coverage to match market readiness. *Space environment:* Debris (including from other operators) poses collision risk. International pressure to manage space traffic is growing; operators mitigate by autonomous avoidance and sharing tracking data.

Implementation Roadmap (New Entrant): Below is a high-level timeline/flow of steps to deploy a LEO broadband network:



Each phase involves iteration: for example, early test launches inform final constellation design; initial market feedback can refine pricing. Partnering with launch providers early (e.g. contracting mass launches by Year 3) is crucial to hit deployment milestones. Ground infrastructure (gateways, NOC, customer support) should be built in parallel with constellation growth, scaling capacity as satellite service becomes available.

Key Open Questions & Next Steps: Uncertainties remain, including: What is the true market demand and acceptable pricing in each region? (Pilot programs or surveys can help answer.) How will spectrum sharing evolve, especially above 30 GHz? (Following WRC-23 outcomes and national consultations is needed.) Can satellite life/cost be improved (e.g. cheaper photonics or reusable sats)? What network architecture best balances satellite vs gateway routing? (Experimental networks or simulations using prototypes may help.) Next steps include detailed technical trade studies (beamforming architectures, link budgets), regulatory engagement (ITU filings, drafting license applications), and financial modeling under various user-growth scenarios. Engaging potential customers (governments, telcos) early can secure commitments to ensure revenue once service is live.

Tables & Figures: The report above includes Table 1 comparing GEO/MEO/LEO and illustrating approximate CAPEX/OPEX. Additional tables could compare vendor options (e.g. satellite bus suppliers, launcher capacities) and international regulatory regimes (spectrum bands, license requirements) if more detailed data is available. A network diagram (satellite-ground mesh) and deployment flowcharts

(e.g. Gantt timeline above) aid understanding. Sources include official filings (e.g. FCC authorizations 【39†L129-L137】), industry whitepapers, and recent academic analyses 【17†L181-L189】 【56†L490-L499】 to ensure up-to-date, credible information.

References: Key references used include regulatory documents (ITU WRC-19 outcomes 【39†L129-L137】), academic techno-economic studies 【17†L181-L189】 【56†L490-L499】 【15†L69-L77】 , and industry reports/blogs 【2†L69-L77】 【64†L608-L616】 【70†L92-L100】 . These are cited inline above.
